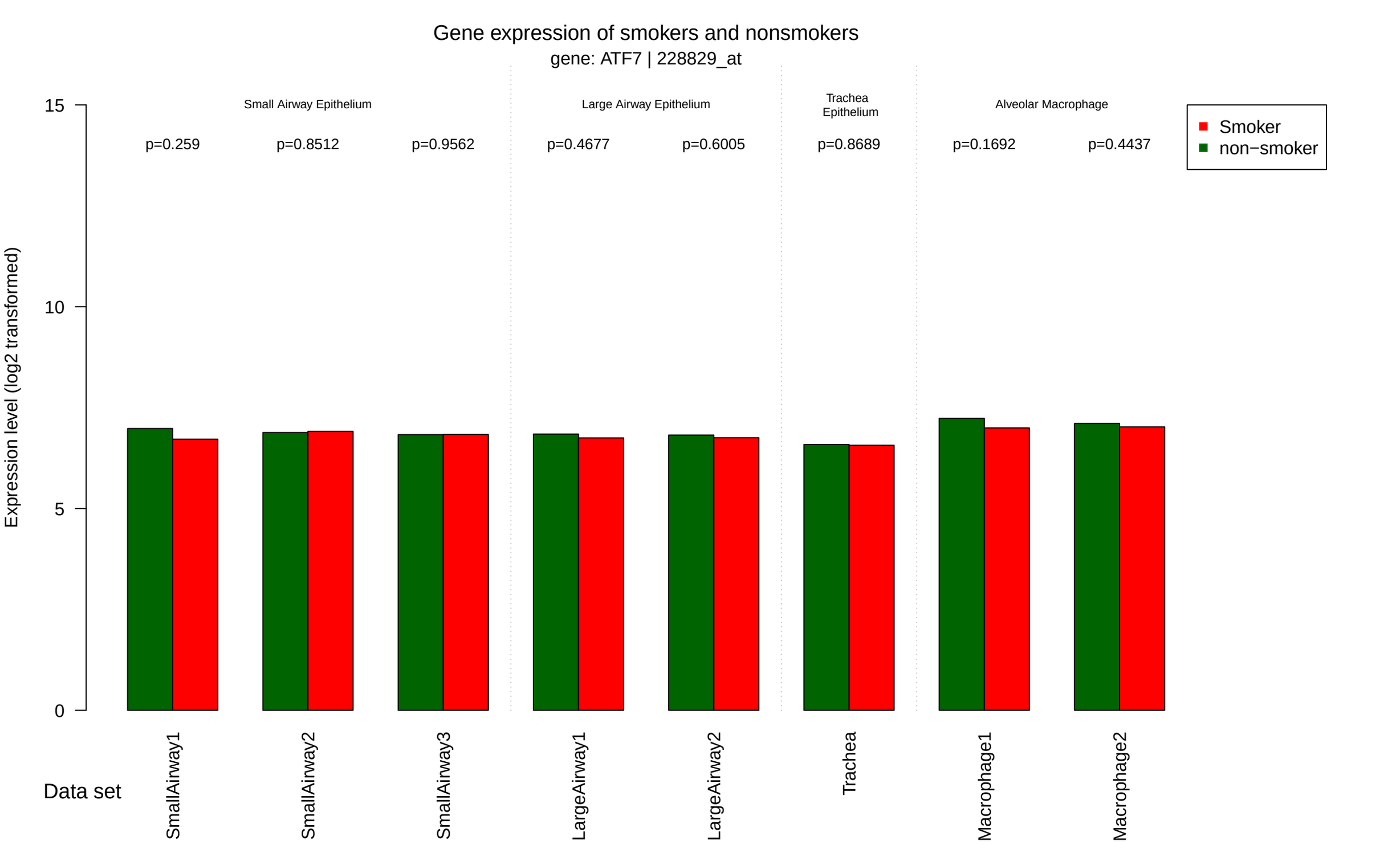
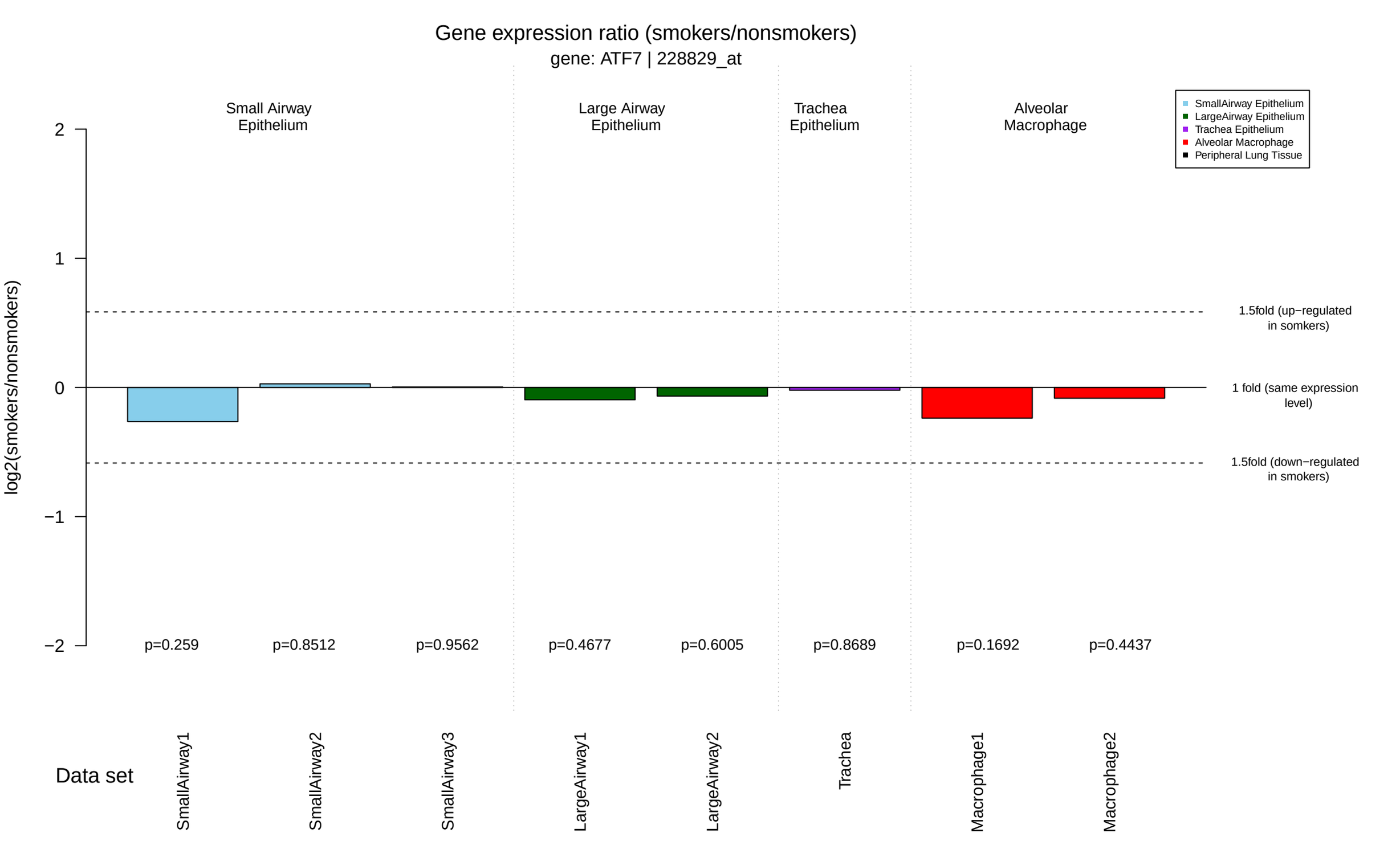
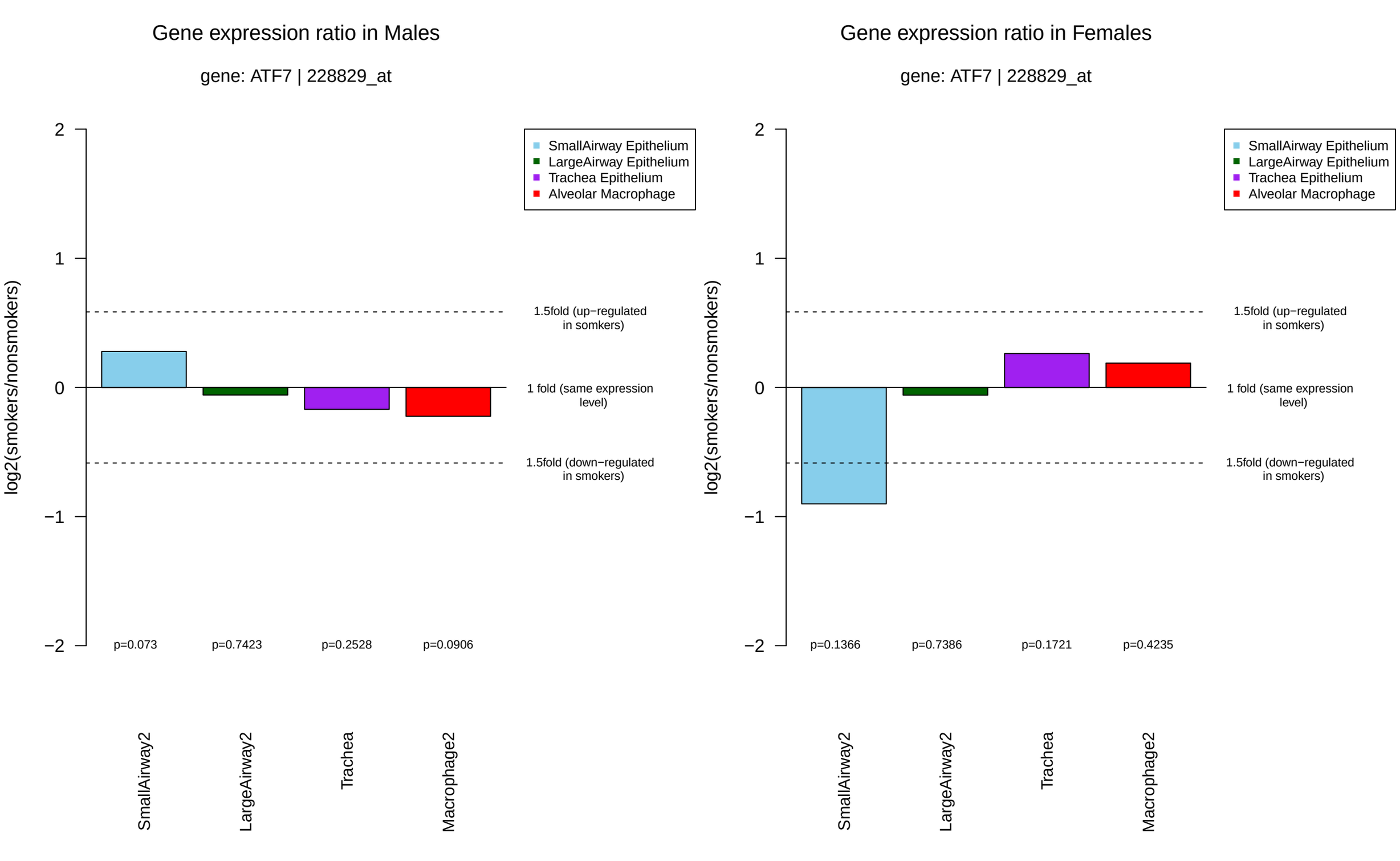
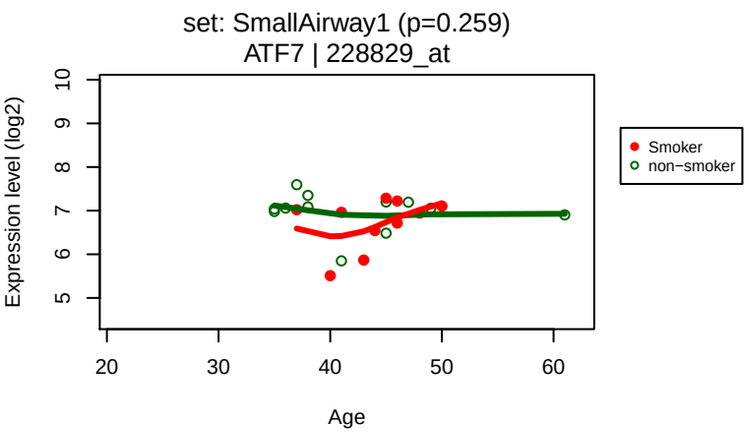
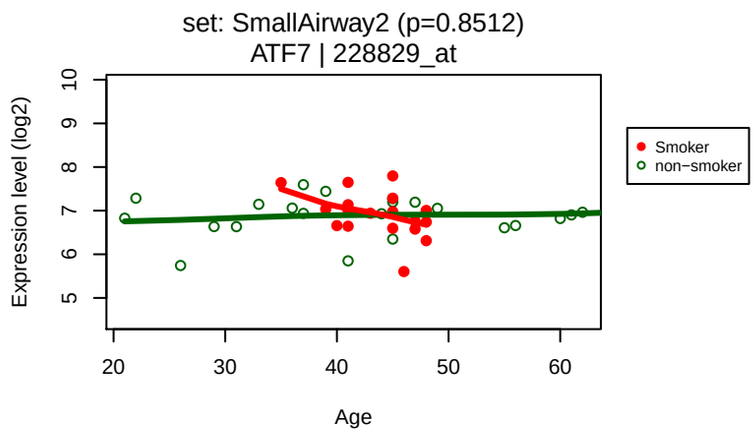
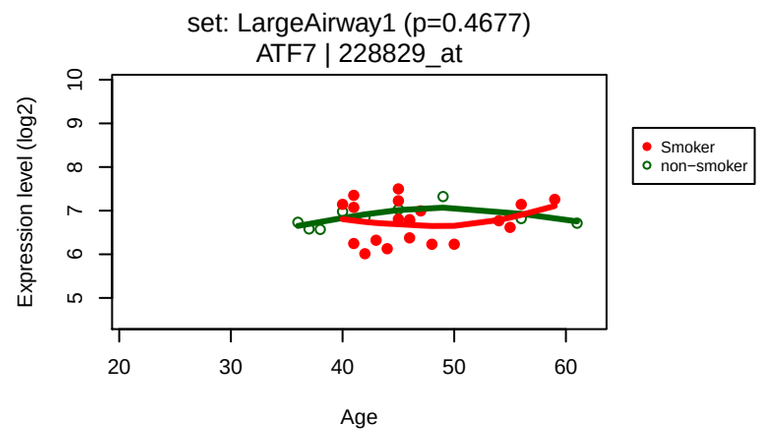

Gene expression of smokers and nonsmokers
gene: ATF7 | 228829_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.259
p=0.8512
p=0.9562
p=0.4677
p=0.6005
p=0.8689
p=0.1692
p=0.4437
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

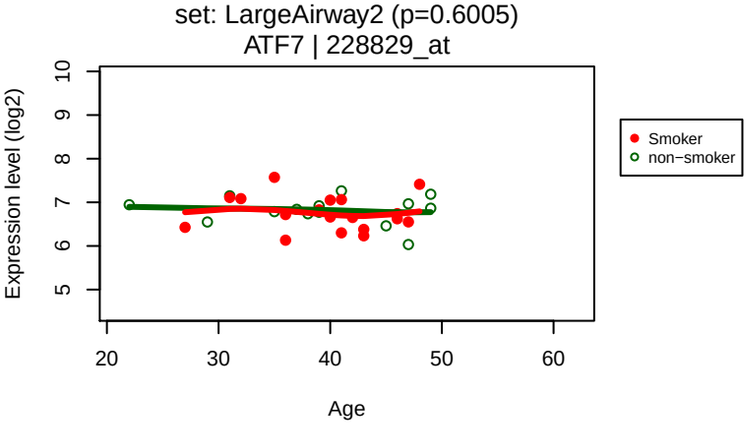

Gene expression ratio (smokers/nonsmokers)
gene: ATF7 | 228829_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.259
p=0.8512
p=0.9562
p=0.4677
p=0.6005
p=0.8689
p=0.1692
p=0.4437
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

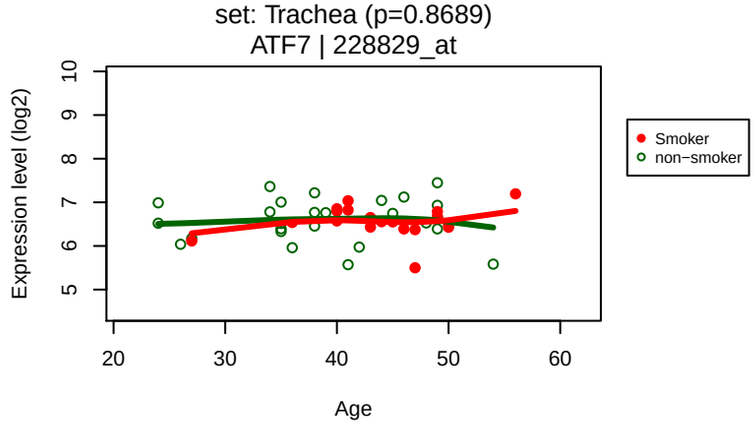

Gene expression ratio in Males
gene: ATF7 | 228829_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.073
p=0.7423
p=0.2528
p=0.0906
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF7 | 228829_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1366
p=0.7386
p=0.1721
p=0.4235
SmallAirway2
LargeAirway2
Trachea
Macrophage2

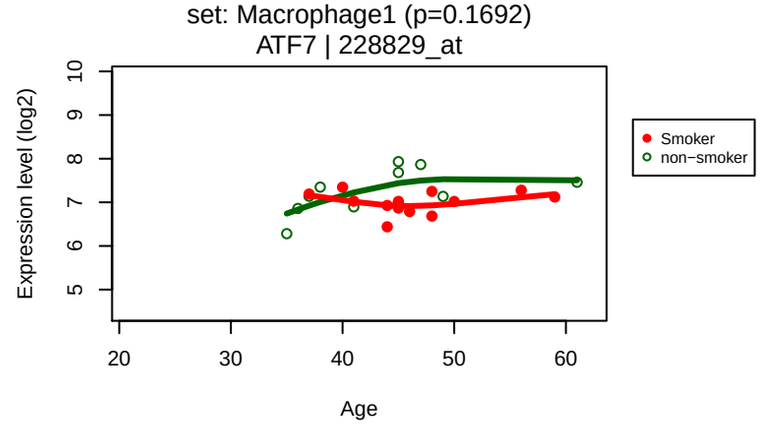

set: SmallAirway1 (p=0.259)
ATF7 | 228829_at
Expression level (log2)
Age
Smoker
non−smoker

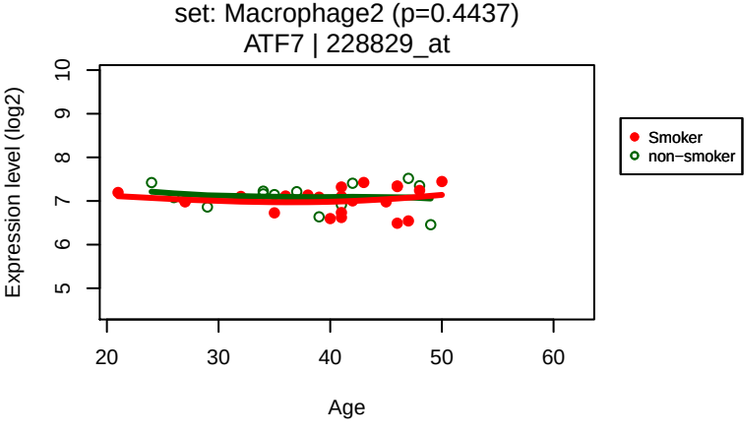

set: SmallAirway2 (p=0.8512)
ATF7 | 228829_at
Expression level (log2)
Age
Smoker
non−smoker

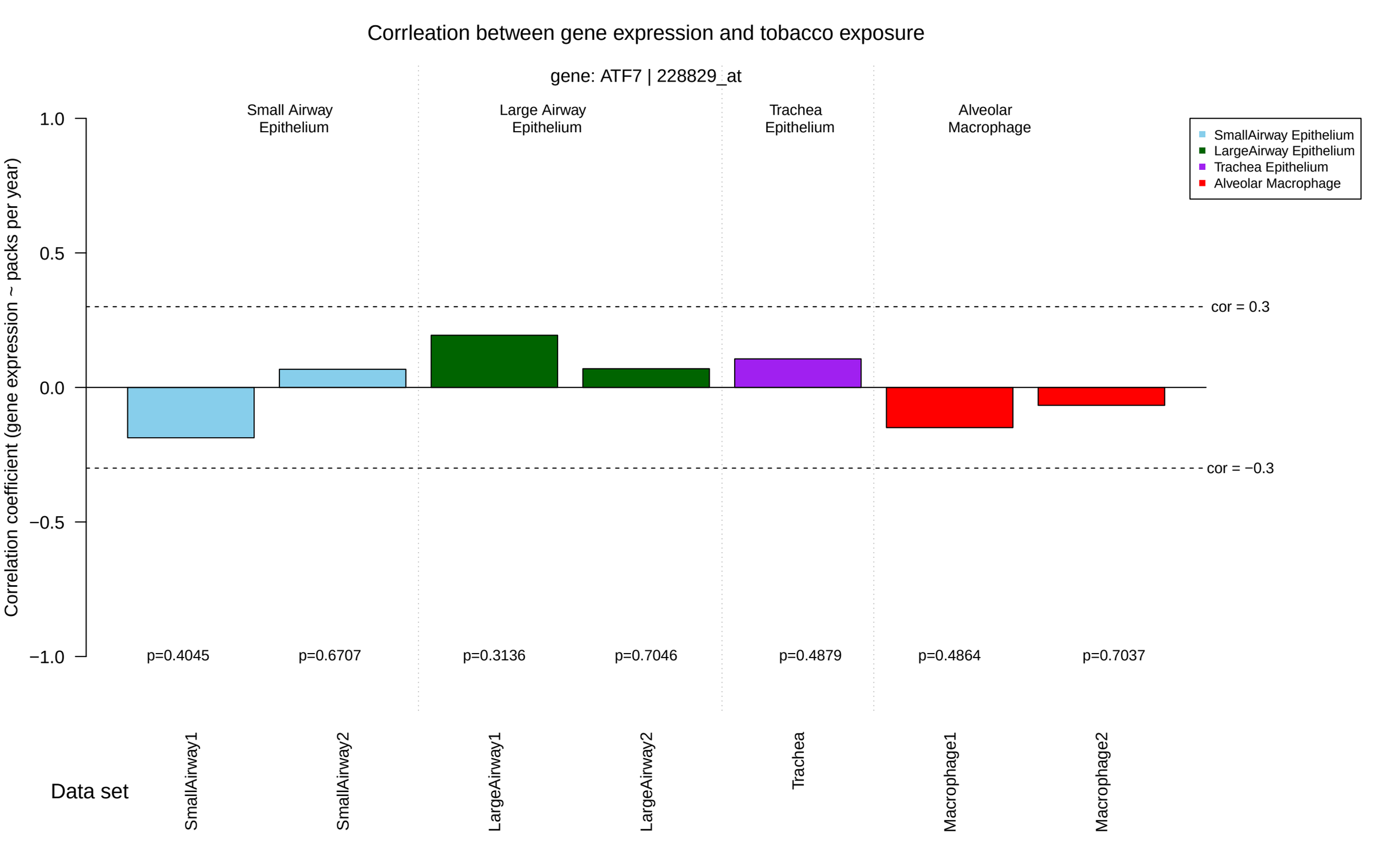

set: LargeAirway1 (p=0.4677)
ATF7 | 228829_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6005)
ATF7 | 228829_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8689)
ATF7 | 228829_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1692)
ATF7 | 228829_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4437)
ATF7 | 228829_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF7 | 228829_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4045
p=0.6707
p=0.3136
p=0.7046
p=0.4879
p=0.4864
p=0.7037
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2